



Rafael Coimbra Azeiteiro

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About Me

Aerospace Engineering MSc student at Instituto Superior Técnico, Lisbon University, Portugal.

I am highly motivated, adaptable, and thrive in both team-based and independent work. Through my academic journey, I have collaborated with international students, which has strengthened my ability to work across cultures.

Master's dissertation research on Guidance, Navigation and Control (GNC) for Martian atmospheric entry, with emphasis on high-enthalpy entry dynamics, flight mechanics and closed-loop guidance.

Education

Instituto Superior Técnico, Lisbon, Portugal - MSc in Aerospace Engineering Sept 2024 - Today

- Major: Space ([Curriculum](#))
- Minor: Aerodynamics and Propulsion

Instituto Superior Técnico, Lisbon, Portugal - BSc in Aerospace Engineering Sept 2021 - Jul 2024

- Cumulative Grade: 16/20 ([Curriculum](#))

Externato Ribadouro, Porto, Portugal - Highschool Sept 2018 - Jul 2021

- Course: Science and Technology
- Cumulative Grade: 19.6/20

Projects

Member of [North Space](#) - 2026

- **Astra Team Member, R&D Branch (2026-Now)** - Contributing to the development of an N_2O /ethanol liquid propulsion system for integration into a European Rocketry Challenge (EUROC) competition rocket.

[ISTrain](#) (IMechE Railway Challenge) - 2025

- **Technical Director (2025-2026)** - Leading a 30+ member railway student team in Instituto Superior Técnico. Responsible for strategy, system integration, sponsor relations and team engineering deliverables.
- **Vehicle Dynamics Team Lead (2025)** - Led a multidisciplinary subteam into the design and validation of the train's suspension system and pneumatic-actuated braking architecture.
- **Powertrain & Energy Systems Engineer (2025)** - Developed and optimized the electric powertrain and cooling systems, applying principles of energy storage, vector control, and passive cooling strategies.

Member of [Aerotéc](#) (Aerospace Engineering Students Group) - 2022

- **GTW - Website (2023-2024)** - Coordinated communication across Aerotec's website working groups.
- **Website & Technologies Lead, Aerospace Week (2023)** - Organized Aerospace Week 2023 and managed the 2024 website ([Link](#)), applying Kanban for project efficiency.

- **Rocket Workshop Member (2022-2023)** - Delivered workshops on rocket stability and design, guiding students through building and launching model rockets.

Experience

Summer Camp Counselor, Centro De Férias Desportivas Nos Mochos, Espinho, Portugal Jul 2024 - Sept 2024

- Summer Camp Counselor Volunteer at "Centro De Férias Desportivas Nos Mochos", dedicated to creating a fun, safe, and enriching environment for all campers aged from 6 to 15 years old.

Volunteer, Cruz Vermelha, Espinho, Portugal Since 2022

- Collected food donations from major chains to support poverty relief efforts.

Core Competencies

- Proficiency in C, Python, Matlab, Simulink.
- Multidisciplinary optimization (OpenMDAO), Aerostructural Optimization (OpenStruct).
- CAD (SOLIDWORKS), CFD (Star CCM+, Ansys, OpenFOAM), Pneumatics (Festo FluidSIM), EM Analysis (CST Studio Suite, LTSpice, KiCAD).
- Frontend and Backend Web Development, NoSQL (MongoDB), JSON, Node.js.
- Applied Requirements Engineering, Agile and Kanban Methodology.
- MS Office, LaTeX.

Soft Skills

 **Communication:** Public Speaking; Active Listening;

 **Collaboration:** Teamwork; Multidisciplinary Coordination; Conflict Resolution

 **Problem Solving:** Analytical Thinking; Root-Cause Analysis; Decision Making

 **Ways of Working:** Adaptability; Growth Mindset

Languages

- Portuguese - Native
- English - C1 (CAE, Grade B, Cambridge)
- Spanish - B1 (Good reading comprehension)